



(a) 1D Random LLaMA2-7B-chat. There exists certain perturbed models that are significantly safer than the original aligned model.

(b) 1D Random Vicuna-7B-v1.5. Replacing the default Vicuna system prompt with the LLaMA2 system prompt improves the overall safety in the model’s local region.

Figure 4: When evaluating the safety landscape using jailbreaking queries, we find that these queries are highly sensitive to perturbations in model weights.

aligned model and multiple perturbed models in its local region. These observations from our safety landscape research provide new insights for future work on LLM attacks and defenses.

7 Safety vs. Capability Landscape

We evaluate on three datasets covering capabilities in math, history, and policy from MMLU [16]. The shape of the LLM capability landscape is drastically different from the one in the LLM safety landscape; these landscapes do not exhibit the same trend, further confirming that the basin shape is indeed unique to the safety of LLM. We provide a detailed analysis in Appendix C.

8 Conclusion

We discover a new phenomenon observed universally in the model parameter space of popular open-source LLMs, termed as “safety basin”. Our discovery inspires us to propose the new VISAGE safety metric that measures the safety in LLM finetuning by probing its safety landscape. Visualizing the safety landscape of the aligned model enables us to understand how finetuning compromises safety by dragging the model away from the safety basin. LLM safety landscape also highlights the system prompt’s critical role in protecting a model, and that such protection transfers to its perturbed variants within the safety basin. These observations from our safety landscape research provide new insights for future work on LLM safety community.

Limitations and Future Work

We believe there are multiple directions for future research, and our work is an important first step in exploring the safety landscape of popular open-source LLMs. Given our findings that the shape of the LLM capability landscape differs significantly from that of the LLM safety landscape, a potential direction for future work is to explore how to better balance the tradeoff between capability and safety, *e.g.*, finding the optimal capability performance for a given dataset while staying within the safety basin. Another direction is proposing additional sub-metrics such as basin width, depth, and smoothness. Our VISAGE score, defined as the average safety margin of all models sampled along random directions, can be considered as an average depth within the safety basin. The VISAGE score is a byproduct of our novel findings on the safety basin, and we hope our study will inspire further research into proposing more metrics, including width and smoothness.

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